RUNGTA INTERNATIONAL SKILLS UNIVERSITY
SCHOOL OF INFORMATION TECHNOLOGY

BACHELOR OF COMPUTER APPLICATION

<table>
<tr><td>PS No.</td><td>5</td></tr>
<tr><td>Project Title</td><td>Sales Data Analyzer</td></tr>
<tr><td>Problem Statement</td><td>Analyse a sales CSV file to find total revenue, best-selling products and region-wise performance using grouping, and visualise the results.
Language: Python 3 Concepts: Pandas, CSV, Matplotlib, Grouping
Note: CSV columns: Date, Product, Region, Quantity, Amount.</td></tr>
<tr><td>Test Cases / Case Statements (4 cases)</td><td>
<table>
<tr><th>INPUT</th><th>EXPECTED OUTPUT</th></tr>
<tr><td colspan="2">Test Case 1: Normal Input</td></tr>
<tr><td>sales.csv:
2026-08-01,Laptop,North,2,100000
2026-08-02,Mouse,South,10,5000
2026-08-03,Laptop,South,1,50000
Menu option 1 (Summary)</td><td>Total Revenue: 155000 | Transactions: 3 | Units sold: 13</td></tr>
<tr><td colspan="2">Test Case 2: Boundary / Minimum Input</td></tr>
<tr><td>sales.csv: 1 row only
2026-08-01,Mouse,North,1,500
Menu option 1 (Summary)</td><td>Total Revenue: 500 | Transactions: 1 | Top product: Mouse</td></tr>
<tr><td colspan="2">Test Case 3: Invalid Input</td></tr>
<tr><td>sales.csv:
2026-08-01,Laptop,North,2,100000
2026-08-02,Mouse,South,10,abc
2026-08-03,Laptop,South,1,50000</td><td>Row 2 skipped: invalid Amount 'abc'
Total Revenue: 150000 | Transactions: 2</td></tr>
<tr><td colspan="2">Test Case 4: Different Valid Scenario</td></tr>
<tr><td>Same CSV as Test Case 1
Menu option 2 (Group by Product & Region)</td><td>Laptop: 150000 | Mouse: 5000
North: 100000 | South: 55000
Bar chart 'Region-wise Sales' displayed</td></tr>
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<tr><td>Bare Minimum Requirements</td><td>1. Write the solution in Python 3 (one .py file) and run it from the console.
2. Read the CSV file using Pandas and clean invalid or duplicate rows.
3. Use groupby() to find product-wise and region-wise sales.
4. Show the top 5 selling products.
5. Plot at least two Matplotlib charts (bar and line).</td></tr>
<tr><td>Instruction</td><td>1. Understand the problem statement and define the problem clearly.
2. Develop the solution using proper logic and programming concepts.
3. Implement all required features and pass the given test cases.
4. Test and debug the project to ensure correct output.
5. Prepare the documentation/report and upload the code and report to GitHub.
6. Complete and submit the project within the given deadline.</td></tr>
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